HARVARD | BUSINESS | SCHOOL

9-910-045
REV. MAY 13, 2013

JOHN S. HAMMOND

C.K. Claridge, Inc.

On a Sunday in mid-September 2009, Christine Schilling was in the office of Ralph Purcell, president of C. K. Claridge, Inc. (CKC). Schilling, recently hired by Purcell, was going over an analysis she had recently prepared and discussed at a meeting in New York with the firm's intellectual property attorneys. Purcell hoped that by the end of the afternoon, aided by Schilling's insights, he would be able to establish a course of action that might hasten the final settlement of a patent suit brought against CKC three years earlier by the Tolemite Corporation and its licensee, Barton Research and Development (BARD).

The Contenders

CKC was founded in Milwaukee, Wisconsin, in 1948 as a commercial outlet for the inventive genius of Dr. Charles K. Claridge, an astute organic chemist. Dr. Claridge owned and managed the company until 1996, when, desiring to retire, he sold it along with all of its patents and products to Arnoux Industries, a small Chicago-based conglomerate. CKC continued to prosper as an Arnoux subsidiary and by 2009 had projected annual sales of about $105 million, 14% of the Arnoux total. About 10% of CKC's sales in 2009 were derived from a chemical component called Varacil, whose manufacturing process was the subject of the patent suit. The remainder of its sales included a wide range of specialty organic chemical products, sold in relatively small volume, primarily to the pharmaceutical industry.

Tolemite, also headquartered in Chicago, was a chemical and pharmaceutical manufacturer with estimated 2009 sales in excess of $3.0 billion. In 2000 Tolemite had been awarded a 17-year patent covering various aspects of a new, low-cost process for synthesizing Varacil. The techniques covered by the patent had been discovered at Tolemite's research facility in 1994 as an offshoot of another project. (Tolemite had filed a patent application in early 1995.) Since Tolemite was neither a user nor a producer of Varacil, it had decided to offer the use of the patent, under license, to BARD, the principal Varacil producer in the United States.

BARD, located in Evanston, Illinois, had begun as a small research company. By 2000, however, it had dropped all research and was involved solely in the production of Varacil. To maintain its position as industry leader, BARD had accepted Tolemite's licensing offer in 2000 and had converted all Varacil production to the new process, thereby replacing all of its natural Varacil production with

---

Senior Lecturer Dr. John S. Hammond III and Research Associate Donald L. Wallace prepared the original version of this case, "C.K. Coolidge, Inc. (A)," HBS No. 894-017, which was revised by Professor James K. Sebenius and abridged by Professor Jan Hammond. This version was prepared by Professor James K. Sebenius. Certain details have been disguised. HBS cases are developed solely as the basis for class discussion. Cases are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management.

synthetic Varacil. In return for the use of the patent, BARD had agreed to pay Tolemite a 4% royalty on all BARD's revenues (e.g. gross sales or other economic proceeds) from synthetic Varacil. In addition, BARD had received rights to sublicense to other Varacil producers who became interested in the process and to work out individual (royalty) agreements with producing firms. Under these sublicensing agreements, the first four percentage points of any revenues received by BARD would go to Tolemite, and any excess would accrue to BARD. (For example, if BARD negotiated a 5% sublicense fee, Tolemite would get the first 4% and BARD would get the 1% excess.)

In 2005, five years after Tolemite had received its patent, a research chemist at CKC had, quite independently, discovered a very similar process for synthesizing Varacil. The CKC researchers, however, had not felt that the new processing techniques could be patented. Thus, no patent search had been initiated and production facilities had simply been converted to the new process. At the time, no one at CKC had suspected the degree to which its new process was similar to the one originated by Tolemite and covered by Tolemite's process patent. It was with some surprise then that CKC management learned that it was being sued by Tolemite and BARD for patent infringement.[1]

Varacil

Varacil was a chemical substance sold almost exclusively to pharmaceutical manufacturers. Although it appeared in a variety of drug preparations, it represented only a minor fraction of any one drug. The economics of its manufacture (high fixed and low variable costs plus economies of scale), however, suggested that it be made in long, high-volume runs. Thus the major drug companies themselves were not involved in its preparation.

Before 2000, Varacil had been processed from naturally occurring organic chemicals found in animal tissue. As a result of the high cost of these natural chemicals, the cost of Varacil itself had been relatively high. With the advent of synthetic Varacil, this situation changed dramatically. Variable costs in the manufacture of synthetic Varacil represented only about 15% of sales, so the synthetic soon drove the natural product out of the market.

In 2009 the national market for synthetic Varacil amounted to some $90 million in sales. On a unit (pounds sold) basis, this market had been relatively stable for several years. As drugs requiring Varacil had been phased out, new ones requiring the compound had typically appeared. Furthermore, industry leaders had no reason to believe that this stability would be lost over the next several years; they projected industry unit sales to be quite flat as far as five to ten years out.

On the dollar value side, however, the story was quite different. Prices for Varacil had been in decline for several years. When converting to the synthetic process, each competitor in the industry had tooled up to supply an optimistic share of the market. Then, when market share objectives were not met, prices were slashed in an attempt to keep manufacturing facilities operating at efficient levels and to bring in as much contribution as possible toward fixed costs. This situation was expected to continue for at least five years. **Exhibit 1** shows industry unit and dollar sales of synthetic Varacil for the period 2000-2009, as well as management's projections for 2010-2020.

[1] Upon filing of the suit, CKC's process engineers had tried hard to develop a process modification that avoided infringing the patent in the manner that the CKC-discovered process allegedly did, but after a year of trying determined that this was not feasible.

In 2009 there were seven principal competitors in the synthetic Varacil market. BARD, with $60 million in sales, took roughly 66% of the market. CKC, with about $10 million in sales, was the second largest and held an 11% share. The remaining five competitors, none of whose Varacil sales exceeded $6 million, then constituted the remaining 23% of the market. By 2006 all seven of the principal competitors were manufacturing synthetic Varacil by nearly identical processes. Only BARD, however, was paying royalties to Tolemite. Based on the logic of the lawsuit, the other producers were technically infringing the patent.

Background on the Litigation

On June 12, 2006, Tolemite (as the patent owner) and BARD (with sublicensing rights) had jointly filed suit in the United States District Court for the Eastern District of Wisconsin charging CKC with having infringed on Tolemite's patent. To remedy the infringement, Tolemite and BARD were seeking an injunction against future infringement over the ten years remaining on the 17-year life of the patent, as well as damages equivalent to license royalty payments of 20% on Claridge's past and future sales of synthetic Varacil.

When confronted with the suit, Purcell had immediately discussed the matter with Aaron Mantiris, general counsel for Arnoux Industries. Both had felt there was considerable evidence indicating that Tolemite's process might not be patentable. At Mantiris' suggestion, CKC had retained the services of Evans and Blaylock, a well-known and highly reputable firm of intellectual property attorneys in New York. These attorneys quickly agreed with Mantiris on the potential weakness of the Tolemite suit. Thus, in 2006, Evans and Blaylock had begun to intensively prepare the case for CKC's defense.

Tolemite's patent contained 12 claims of originality. Like all successful patent applicants, Tolemite had had to demonstrate to the patent examiners that there was no "prior art," e.g. previous patents, applied-for patents, or processes in the public domain—unpatentable, but generally known—that were similar.

In fact, there were many instances of patents being successfully challenged. However, once a patent was issued, the burden lay with a potential infringer to prove that the claimed invention was unpatentable. And, unless there was a serious irregularity such as fraud associated with its issuance, a patent was treated as valid unless and until it was legally invalidated. In the matter of synthetic Varacil, Mantiris argued that Tolemite had not, in fact, introduced any novelty, but had merely observed and harnessed a naturally occurring process that, in itself, was not patentable.

A patent holder whose patent was infringed was entitled to sue the infringer for damages to compensate for sales and profit wrongfully obtained. In awarding damages in a case like that brought jointly by BARD and Tolemite against CKC, a court would almost certainly consider a reasonable royalty and the plaintiff's lost profits—with the award potentially trebled in the case of egregious circumstances, which seemed unlikely in this suit. In determining the amount to be demanded in a lawsuit, the plaintiff usually calculated these damages in a way most favorable to itself. However, if the plaintiff prevailed in court, the actual damages awarded were often considerably less. In the Varacil matter it was the strongly held opinion of both Mantiris and the Evans and Blaylock lawyers—based on considerable comparable experience, relevant legal precedents, and other data--that although the suit specified a 20% royalty payment, the amount awarded if CKC lost the suit would be equivalent to approximately 10% royalties on past and future sales through the life of the patent. Purcell was persuaded of the accuracy of this assessment; as was common, in the view of CKC and its attorneys, the plaintiffs had claimed an amount roughly double their likely award--if they ended up prevailing in court.

From 2006 to 2009 a partner in Evans and Blaylock supervised the firm's extensive defense work in liaison with Mantiris. Frankly, CKC management preferred to ignore this unpleasant possibility, focusing instead on their real business of Varacil production and sales. Indeed Purcell and his senior colleagues considered the suit to be an expensive nuisance and were content to drag their feet in the hope that BARD and Tolemite's case might simply collapse. Late in 2008, however, a tentative trial date was set for January 2009. Before a firm date could be set, Purcell and Mantiris decided, with the concurrence of the patent attorneys, to make at least a token effort at a pretrial settlement. Their offer amounted to the payment of all past and future liabilities at a royalty rate of 2.5% of sales. Tolemite and BARD immediately rejected this offer. Eventually, the case reached the court docket and a trial date in October 2009 was set.

By September, Purcell was becoming extremely uneasy over the high—and increasing—level of attorneys' fees, which by this point included extensive research involving the U.S. Patent Office, detailed reviews and assessment of potentially relevant statute and case law, as well as the expenses of various independent experts and consultants, some of whom had performed costly technical studies that might be useful in CKC's defense. Almost unnoticed as they were incurred, these fees and expenses had already reached a total of almost $2 million and, if the trial were to take place as scheduled, they would surely loom large in comparison with the total value of any successful defense. Furthermore, unlike in some jurisdictions in which the "loser" of a court suit pays the "winner's" legal fees, these legal fees and any future ones would not be recoverable, even if CKC won its case against BARD and Tolemite.

In response to this uneasiness about both the progress of the suit and the alarming accumulation of the attorneys' fees and expenses, Purcell decided on two immediate actions. First, through Mantiris he arranged for a meeting in New York City to review the case thoroughly with the patent attorneys. Second, he asked Schilling, his recent hire, to review the case, and he hoped, to bring a fresh viewpoint to bear.

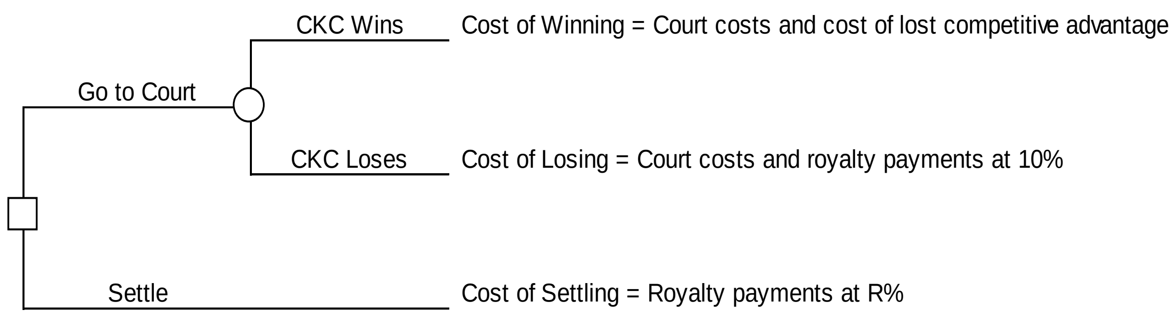
Schilling's Analysis and the Meeting with the Patent Attorneys

Christine Schilling, a recent graduate of the Harvard Business School, had studied the quantitative analysis of decision problems, and thought that a decision analysis approach would help her understand the choices and possible outcomes. She began by pulling together actual and forecast data on CKC and industry production and sales of Varacil, as well as the effects of various possible royalty rates; these data, mainly from trade association sources, were generally accepted within the industry. (See Exhibit 1 for this and other data.) She also checked with CKC's CFO to confirm CKC's cost of capital (10%) to "discount" future revenues/costs, and reviewed information that CKC had learned through the legal process so far. Schilling started with a simplified analysis of CKC's fundamental decision: to settle or fight. She explained her reasoning to Purcell. "The basic 'tree' is quite simple – either CKC goes to court and contests the patent, or we settle for a royalty amount *R*% of past and future sales. And if we go to court, there are eventually only two outcomes – we either win or lose." Schilling summarized her ideas in a decision diagram (**Figure A).**

She continued: "If we lose, we'll have to pay court costs and an estimated 10% royalty payments on our past and future sales of synthetic Varacil over the life of the patent. Winning would be a mixed blessing – we wouldn't have to pay a royalty of course, but then neither would BARD, since the underlying patent basis of its contract with Tolemite would no longer be valid. Right now we enjoy a competitive advantage over BARD because it pays a 4% royalty to Tolemite while we don't; of course, BARD's royalty payments would disappear (going forward) if the court finds the patent to be invalid." (See **Tables A-1 and A-2 in Exhibit 1** for analysis of royalties for CKC, BARD, and others.)

Nodding in agreement, Purcell added, "BARD would certainly regard that as an unexpected gift from us! However, it may not be at all obvious to BARD or others that the high fixed costs and competitive pressures of the Varacil industry will quickly push BARD—if it stops paying the royalty—to lower its prices. To stay in the game, we'd have to reciprocate and drop our prices as well. I'm not sure that others would necessarily analyze things this way, but given my experience and analysis, I'm strongly confident of this judgment about our 'lost competitive advantage' even if we win in court." Schilling concurred.

Figure A Basic Decision Diagram



Although Christine Schilling made a series of notes on the assumptions behind her analysis (see **Exhibit 2**), she took Purcell through the quantitative version of her thinking, initially referencing **Figure A** and the tables in Exhibit 1: "To determine the actual financial cost, consider the outcome that we go to court and win. For that outcome, with the benefit of extremely careful assessments from counsel, we estimate the additional court costs (legal fees, consultant expenses, etc.) of going forward in court to be $1.5 million. From **Table A-1** in Exhibit 1, the present value of the cost of our 'lost competitive advantage' would be $1.71 million. So the "cost of winning" would be the sum of these two quantities--$1.5 million plus $1.71 million--or $3.21 million. The cost of losing would be the (same) additional court costs ($1.5 million) plus the present value of the cost of a 10% royalty on our past and future sales; from **Table A-1**, that would be $6.94 million. This implies that the total cost of going to court and losing would be $8.44 million ($1.5 million plus $6.94 million). If the cost of settling at 10% of past and future sales is $6.94 million, then the cost of settling for one percentage point of past and future sales would be $.694 million, or very roughly $700 thousand. Thus, at R% of past and future sales--where the value of the settlement royalty rate, R, can range from 0 to 10---the total cost of settlement would be $.694R million. For example, if we settled at R=7, a 7% royalty, we'd owe 7(.694) = $4.858 million in present value of past and future sales." She summarized (with all figures in millions of dollars):

Cost of Winning = 1.5 + 1.71 = 3.21

Cost of Losing + 1.5 + 6.94 = 8.44

Cost of Settling for R percent of past and future sales= .694R.

Ralph was on board with this analysis, but asked Christine the obvious question: "This is all fine, but it doesn't take into account the probabilities of winning and losing, which matter a lot."

Christine was ready for Ralph's question. "Indeed. But Ralph, I don't think it is particularly meaningful to put a single number on, say, the probability of our winning, plug that into the analysis, and crank it through. Our Evans and Blaylock counsel may have one view; Mantiris may have another. After hearing them, you and I might come to a still different judgment about the probabilities. So let's just call the probability of CKC winning in court "p", without specifying p's value. Of course, we can let the value of p vary from 0 (meaning NO chance of our winning) all the way to 1.0 (meaning CKC is absolutely rock-solid certain to prevail in court). By letting the value of p vary, we'll see how sensitive our conclusions are to different forecasts of our chances in court."

"OK, I'm all in favor of sensitivity analyses, especially for back-of-the-envelope exercises like this, but," Ralph continued, "if p is the value of winning, what's the probability of our ultimately losing?" Pausing a brief moment, then slightly chagrined, he answered his own question: "Sorry, that's obvious. If the p is our subjective probability of CKC winning in court, then of course we lose with probability 1-p. If we have a 70% chance of winning, then we have a 30% chance of losing."

Schilling picked up the argument: "As an economic matter, we should only settle if the cost of settling for a given assessment of our chances in court (p)—meaning the highest value of R which we'd offer to BARD and Tolemite for that assessed value of p--is LESS than the expected cost of going to court (provided we aren't either risk-averse or risk-prone)." Ralph nodded his head twice, that yes, 1) this logic was correct and that yes, 2) for this scale of decision, CKC should unemotionally play the odds as best they judged them, meaning CKC should make decisions on an expected value, risk neutral basis."

So Christine put the various factors together as follows: "We should settle for R% of past and future sales provided that the expected cost of settling the suit at R% is less than the expected cost of going to court. Symbolically,

[Expected cost of settling at R%] < [Cost to CKC of winning in court] x [probability of CKC winning] + [Cost to CKC of losing in court] x [probability of CKC losing]

or, with numbers (in millions of dollars)

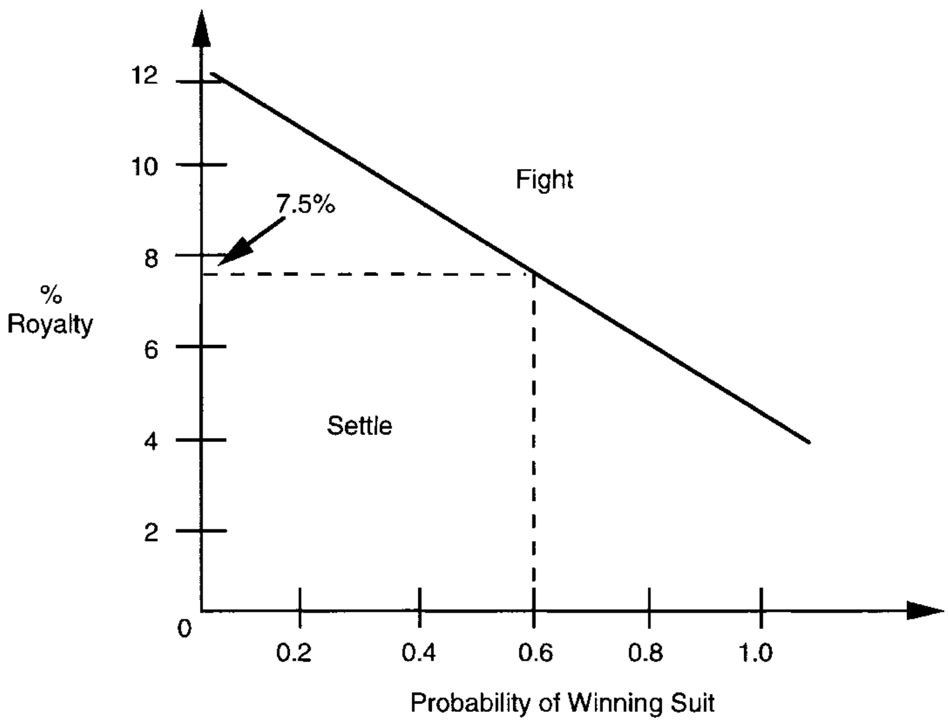
$$.694R < [1.71 + 1.5]p + [6.94 + 1.5][1-p]$$

Solving this equation for the royalty rate R in terms of the probability of winning p, we get that we should settle as long as $R < 12.16 – 7.54p$.

Ralph responded: "So let's play out some of the implications of this. We've got to make sure that we're using the analysis to inform our judgment, not replace it, or get hypnotized by equations that are only as good as the assumptions behind them. Say that CKC was absolutely sure to win--that would mean we were 100% positive of a court victory, or that p=1. Plugging p=1 into this expression, we should still settle for amounts up to R < 4.6, that is, for a royalty rate equal to 4.6%--at least on a purely economic basis. Wow! If you analyze it, I guess it is actually worth a lot to avoid the legal costs and other consequences of going to court, even if we're confident in our case. By contrast, say we were total pessimists and absolutely sure to lose. Then p=0, which we can plug into that expression and get that we should settle for anything up to 12.16%. So, no matter how we see our chances in court, it would seem that there is a lot of room for to negotiate a profitable settlement relative to the litigation alternative."

Schilling broke in "And for any probability of winning, I've graphed the maximum royalty for which we should settle. Take a look at **Figure B**, which is a kind of breakeven line (or "curve"). For example, you can directly read off this graph: if we were 60% sure of a victory (i.e., p=.6 on the horizontal or x-axis), it would still be economically worth settling for up to roughly a 7.5% royalty rate (on the vertical or y-axis). (We could use the equation to calculate more precisely, but all this is pretty loose.) But looking at this breakeven line, for any probability p that we assess of winning in court, we should be willing to settle for any royalty offer below the line, and reject any proposed royalty offers (R) that are above the line (at that value of p)."

Figure B Break-even Curve



Schilling's principal conclusion was that unless the odds of winning the suit were extremely good, any reasonable pretrial settlement was preferable to paying the additional costs and taking a chance on going to court. Purcell, a chemical engineer, was himself well attuned to quantitative analysis and, in fact, liked to support his own arguments with numerical data whenever possible. Intrigued by Schilling's presentation, he invited her to join him and Mantiris on the trip to New York City to meet with the patent attorneys. At that meeting Purcell intended to discuss Schilling's analysis with the lawyers and seek their opinion on the benefits of going to trial.

In New York, the patent attorneys began the meeting by presenting an outline of their case. Everyone attending agreed that the case was indeed a strong one with a high probability of success in the trial phase. The attorneys demurred, however, when asked to give a precise figure for their probability of eventual success in court. At that point, Purcell sketched out Schilling's analysis. He then asked the patent attorneys if they still felt that their probability of success was high enough to merit going to trial. The attorneys were visibly uncomfortable with Schilling's approach. Although they remained convinced of the merits of their case, they agreed that some rethinking was probably necessary before proceeding to trial.

On the way home, Shilling and Purcell discussed the meeting and next steps. Purcell asked whether the decision tree they had used was “too simple” for the reality of the situation CKC faced. Schilling responded, “Now in reality, the tree is a bit more complex if we go to court. The losing party might choose to appeal after learning the outcome of the original trial; and the appeal could either overturn the trial outcome or not.” She sketched out a more complete tree that reflected the possibility of appeal after the original trial, and included possibilities of winning or losing the appeal. (See **Exhibit 3.**). She then took Purcell through the various possibilities in this more detailed scenario.

“I could develop some good estimates of the likelihood that the losing party will choose to appeal and of the likelihood of winning the appeals, as well as the legal costs associated with an appeal. For example,” Schilling continued, “if CKC loses the original trial and appeals that decision, the lawyers tell me that the added cost of this extra legal phase would be another $1.5 million. However, if CKC wins the original court fight and BARD/Tolemite appeal the decision, the appeal will cost us less, likely about $750,000. An appeal seems very likely (perhaps 90%) no matter who wins. Using these figures, I could go through a similar process to the simpler analysis I first did in order to determine for any given out-of-court settlement offer *R*. ”

Purcell leaned back and paused for a moment, “Christine, this is a real contribution. I really like the clarity of your analysis. It lays out all the factors in logical relationship to our key decision. I really wish we’d done this before spending all the money we have so far on legal fees and expenses. But before we push this line of analysis much farther, or get too lost in the numbers, let’s really think through the implications for our approach going forward. Can you meet at the office Sunday afternoon? It will be quiet so we can think through the analysis and elaborate it if necessary, in order to figure out the best possible strategy for dealing with this lawsuit.

Exhibit 1 Unit and Dollar Sales of Synthetic Varacil by Company Plus Analysis of Potential Royalties

	BARD		Claridge		All Others	
Year	**Thousands of Lbs.**	**Thousands of $**	**Thousands of Lbs.**	**Thousands of $**	**Thousands of Lbs.**	**Thousands of $**
2000	1	1,530	0	0	0	0
2001	5	7,380	0	0	0	0
2002	20	26,760	0	0	0	0
2003	60	65,690	0	0	0	0
2004	68	80,220	0	0	0	0
2005	76	90,450	0	0	0	0
2006	83	96,240	1	1,110	0	0
2007	89	95,460	6	5,760	2	2,130
2008	94	78,990	11	9,360	19	16,080
2009	100	60,000	17	10,500	35	21,000
2010	100	60,000	17	10,200	35	21,000
2011	100	57,000	17	9,690	35	19,950
2012	100	54,000	17	9,180	35	18,900
2013	100	48,000	17	8,160	35	16,800
2014	100	45,000	17	7,650	35	15,750
2105	100	45,000	17	7,650	35	15,750
2016	100	45,000	17	7,650	35	15,750
2017	100	45,000	17	7,650	35	1,5750
2018	100	45,000	17	7,650	35	15,750
2019	100	45,000	17	7,650	35	15,750
2020	100	45,000	17	7,650	35	15,750

Source: C. K. Claridge, Inc.

Notes: (1) Total unit sales of Varacil (including the natural product) were roughly 150,000 lbs. annually for the period 2000-2009.

(2) In the table above, sales beyond 2009 were projected.

Table A-1 Present Value of Royalties and CKC's Lost Competitive Advantage ($000s)

				Cost of 10% Royalty Payments & Past Liability Claims		Cost of 4% Lost Competitive Advantage[a]		Cost of Settlement at 8% Royalty[b]	
Year	Calendar Year(s)	Discount Factor[c]	Sales	Royalty (10%)	Present Value[d]	Lost Comp. Adv. (4%)	NPV	Royalty (8%)	NPV
0	2006-9	1.000	$26,730[e]	$2,673	$2,673	-	-	$2,140	$2,140
1	2010	0.909	10,200	1,020	930	$410	370	820	740
2	2011	0.826	9,690	970	800	390	320	780	640
3	2012	0.751	9,180	920	690	370	280	730	550
4	2013	0.683	8,160	820	560	330	220	650	450
5	2014	0.621	7,650	770	480	310	190	610	380
6	2015	0.564	7,650	770	430	310	170	610	350
7	2016	0.513	7,650	770	390	310	160	610	310
					$ 6,944		$1,710		$5,560

[a] Schilling assumed CKC would lose a 4% competitive advantage if the patent were overturned.

[b] The actual rate at which the case might settle is unknown; 8% is used as to illustrate the analysis.

[c] A discount factor, used to find the present value of a cash flow, is calculated as $1/(1+i)^n$ where i is the discount rate and n is the period of time until that cash flow is realized. CKC used a 10% a discount rate. Thus, for example, the discount factor used to find the net present value of a cash flow realized in 2 years is $1/(1.1)^2 = 0.826$. Note that columns in this table may not seem to add correctly because of underlying discounting and rounding. Sums, however, are correct.

[d] The present value of a cash flow is found by multiplying the cash flow by the appropriate discount factor. Thus, for example, the present value of $970,000 in two years is $970,000(0.826) = $801,220.

[e] Total past sales (2006-09) are taken from Exhibit 1.

Table A-2 Present Value/Cost of Royalties ($000) to Various Industry Players[a]

			NPV of Royalty Payments at:		
		NPV of Total Sales ($000)	**4%** ($000)	**6%** ($000)	**10%** ($000)
BARD	**Future Sales (2010 – 2016)**	251,430	10,050	15,090	25,150
CKC	**Future Sales**	42,750	1,710	2,580	4,280
	Past Sales	26,730	1,080	1,590	2,673
	Total	69,480	2,790	4,170	6,944
Others	**Future Sales**	88,020	3,510	5,280	8,760
	Past Sales	39,210	1,560	2,340	3,930
	Total	127,230	5,070	7,620	12,690
CKC & Others	**Future Sales**	130,770	5,220	7,860	13,080
	Past Sales	65,940	2,640	3,960	6,600
	Total	196,710	7,860	11,820	19,680

[a] Note: columns do not always add precisely because of underlying present value calculations and rounding. Sums, however, are correct.

Exhibit 2

Christine Schilling's Notes on Her Analysis

The objective of this analysis is to define, in general terms, the relationship between the future expenses and risks faced in the Bard/Tolemite suit with the cost of an immediate settlement. An attempt has been made to break the overall problem into a number of smaller events and action alternatives and to assess reasonable ranges of event probabilities and consequences; these elements are then related mathematically to obtain a solution.

There can be a substantial advantage in this approach in illuminating the basic issues which generally remained submerged in a single assessment of the entire situation. However, there is a potential danger in quantification and simplification of complex problems—the result is apparently so precise and straightforward that it can be very easily overlooked that this result is no better than the assumptions on which it is based.

Assumptions:

I. Exposure to Liability:

A. Past Liabilities

1. If Bard/Tolemite wins, it will be awarded 10% of CKC's $26,730 million of past sales of synthetic Varacil through 2009 — total liabilities of $2.673 million.

2. BARD/Tolemite will settle past liabilities prior to the trial at the same royalty rate applied in future sales.

B. Future Royalties

1. If BARD/Tolemite wins, it will receive 10% of CKC's future sales of synthetic Varacil throughout the remaining life of the patent.

2. The BARD/Tolemite pretrial settlement royalty requirement is unknown.

II. Future Attorneys' Fees and Court Costs will be $1.5 million

III. Basis for evaluation of costs:

A. C. K. Claridge will continue to produce 17,000 pounds of Varacil per year for the next seven years (the remaining life of the patent).

B. The price/pound for Varacil will erode as expected and produce the total sales shown in **Exhibit 1.**

C. In this industry of high fixed and low variable costs and the resulting severe pressure upon price, CKC currently enjoys a competitive advantage proportional to 4% of sales since BARD pays a 4% royalty and CKC does not. If the patent is overturned, the assumption is that BARD will lower its price rather than simply absorb extra profit. The extent of BARD's use of this advantage and its significance for CKC's profitability is illustrated in **Table A-1**.

D. The cost of capital or "value of money" to the corporation to be used in net present value (NPV) calculations (see **Table A-1 and Table A-2**) is approximately 10%.

Exhibit 3 Decision Diagram for Fuller Description of Events